

PAPER Vertical Stability at High Elongation: Displacement Growth Rates, Control-Coil Authority, and VDE Avoidance for a Compact Spherical-Tokamak Breeder

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Abstract

Elongation buys confinement in a compact machine, and it is not optional: the Hyperion spherical-tokamak breeder holds $I_p = 9.66$ MA at elongation $\kappa = 2.0$ with negative triangularity $\delta = -0.30$, aspect ratio $A = 2.5$ and major radius $R_0 = 1.20$ m. The price of that elongation is a passive axisymmetric vertical instability, and in a spherical tokamak the instability is fast. We pose the vertical-control problem formally, from the massless rigid-displacement force balance through the passive-vessel circuit to a proportional-derivative (PD) feedback loop closed on the poloidal-field (PF) coils, and we solve it on the canonical free-boundary equilibrium reconstructed by a GPU-accelerated Grad-Shafranov solver. A single mechanistic result organises the analysis: the normalised growth rate obeys $\gamma\tau_w = \hat{k}/(1 - \hat{k})$, where $\hat{k}$ is the destabilising-to-wall coupling and $\hat{k} \rightarrow 1$ is the ideal wall limit. At $\kappa = 2.0$ the vessel vertical-mode wall time is $\tau_w = 11.9$ ms and $\gamma\tau_w = 8.05$, so $\hat{k} = 0.890$, $\gamma = 677$ s⁻¹ and the growth time is 1.48 ms — spherical-tokamak-typical (NSTX-class). A PD controller on the PF coils arrests the mode: a direct eigenvalue solve of the closed-loop system places a fast pole at -964 s⁻¹ (faster than the open-loop growth) and a slow resistive-relaxation pole at -14 s⁻¹; the stabilising proportional gain threshold is $K_p^* = 30\,302$ V m⁻¹ and the damping threshold is $K_d^* \approx 1156$. From a 5 mm disturbance the excursion peaks at only 8.1 mm at $t \approx 3.54$ ms and reverses, requiring a PF converter of ~ 49 MVA (10.1 kV, 4.9 kA, 5.0 kA ms⁻¹). The elongation carries headroom: the feedback ceiling $\gamma\tau_w \approx 15$ is reached at $\kappa_{\max} \approx 2.37$, so the operating point sits 19% below the controllable limit, with the converter growing super-linearly ($49 \rightarrow 91$ MVA) toward it. The worst case is bounded: a vertical displacement event terminating in a disruption gives a halo current of ~ 2.9 MA and a peak vertical force of ~ 23 MN, a structural specification rather than a threat. The $\kappa = 2.0$ point is a conservative, controllable choice with quantified margin in stability and actuator authority alike.

Keywords: vertical stability; elongation; feedback control; RZIp model; poloidal-field converter; vertical displacement event; halo current; spherical-tokamak breeder

1 Introduction

1.1 Elongation is the lever, and the lever is unstable

A vertically elongated plasma confines more current and more pressure at fixed toroidal field than a circular one: the poloidal-field energy is distributed over a taller cross-section, the safety factor at fixed I_p is raised, and the accessible normalised pressure β_N climbs roughly with elongation. For a compact, high-power-density spherical tokamak (ST), where the central column crowds the machine and the aspect ratio is deliberately small ($A = 2.5$), elongation is the single most economical route to holding a multi-mega-ampere current in a 1.2 m-major-radius device. The Hyperion breeder therefore runs at $\kappa = 2.0$, $\delta = -0.30$, $I_p = 9.66$ MA, $B_0 = 8$ T on axis and a centrepole peak field of 16.84 T [1, 31].

The physics that makes elongation valuable also makes it dangerous. An up-down-symmetric plasma held by a vertical field whose curvature is set up to elongate the cross-section is unstable to a rigid vertical displacement: a small upward shift produces a radial field that pushes the column further up, an axisymmetric ($n = 0$) ideal-MHD mode with no restoring force from the plasma itself [2, 12, 13]. The conducting vacuum vessel slows the mode to a resistive timescale, but it does not remove it; the plasma must be actively held. This is not a Hyperion-specific liability — every shaped tokamak from DIII-D and JET to NSTX, MAST, TCV and SPARC confronts it [3, 5, 24, 27, 28] — but in an ST the wall is close and the growth is fast, so the control margin must be demonstrated, not assumed.

1.2 The design question

The relevant question is therefore not *whether* the plasma is vertically unstable — it is, by choice — but whether the instability is controllable with margin, and at what actuator cost. Concretely we must answer four quantitative questions. (i) How fast is the open-loop mode, given the as-built vessel? (ii) Does a realisable feedback controller on the PF coils arrest it, and with what gain margin? (iii) How much elongation headroom remains before controllability is lost, and how does the converter rating scale toward that ceiling? (iv) Is the worst case — a vertical displacement event (VDE) terminating in a disruption — survivable as a bounded structural load?

1.3 Prior art

The reduced description of the axisymmetric vertical mode has a long pedigree. Mukhovatov and Shafranov [2] established the equilibrium field-index picture; Lazarus, Lister and Neilson [3] gave the massless-filament model with a resistive shell and a resistive control coil that underlies every operational controller since; the ITER-era community formalised the stability margin of elongated plasmas [4, 5, 18] and the linearised free-boundary response models — CREATE-L [6], CREATE-NL+ [7], and their eddy-current-resolving successors — that map a rigid-mode growth rate onto a distributed vessel geometry. Magnetic control theory for tokamaks is now a mature discipline [8, 9, 10, 11]. On the disruption side, the halo-current and electromagnetic-load database [18, 19, 20, 21] and the ITER mitigation strategy [22, 23] set the structural design basis for the VDE endpoint. Spherical tokamaks specifically run near the ideal wall limit for the vertical mode and have documented sub-millisecond-to-millisecond growth times [24, 25, 26].

1.4 Contribution

This paper (i) derives a single closed mechanistic relation, $\gamma\tau_w = \hat{k}/(1 - \hat{k})$, that ties the ST-typical normalised growth rate at $\kappa = 2.0$ to a physically transparent destabilising-to-wall coupling $\hat{k} = 0.890$ (section 2); (ii) closes a proportional-derivative loop on the PF coils as a reduced two-state RZIp model whose eigenvalues, gain thresholds and converter demand are all reported and reproduced by a direct eigenvalue solve (sections 3, 4); (iii) maps the elongation headroom to the feedback-controllable ceiling $\kappa_{\max} \approx 2.37$ and the super-linear actuator trade (section 5); and (iv) bounds the VDE endpoint as a structural load (section 6). Every number is anchored to the frozen Kronos de-risking register — the PF-converter and vertical-control study, the elongation-limit study, the free-boundary equilibrium, and the disruption-load analysis — and to the canonical equilibrium reconstructed on the NVIDIA GPU HPC campaign (section 3). The companion equilibrium and ideal-MHD paper supplies the vertical-stability flag we quantify here [31]; the companion PF-coil-set paper supplies the coil geometry that provides the control authority [32]; and the companion real-time control suite embeds the controller derived here in the full plant [33].

2 Governing equations and the formal problem

2.1 The rigid-displacement approximation

On the resistive timescale of interest the plasma responds to a vertical perturbation as a rigid current-carrying ring: the internal profiles and the boundary shape are frozen, and the single degree of freedom is the vertical position Z of the current centroid. This is the standard reduced description for the axisymmetric vertical mode [3, 4, 8], valid because the ideal (inertial, Alfvénic) growth time is orders of magnitude shorter than the resistive wall time τ_w , so the plasma sits in instantaneous force balance while the wall currents decay. We work to linear order in Z about the canonical up-down-symmetric equilibrium.

2.2 Destabilising force and the field decay index

Let $B_z(R, Z)$ be the externally applied vertical (equilibrium) field. Its curvature is characterised by the *field decay index*

$$n \equiv -\frac{R_0}{B_{z0}} \left. \frac{\partial B_z}{\partial R} \right|_0, \quad (1)$$

evaluated at the magnetic axis $(R_0, 0)$. Vertical equilibrium of a current ring in an external field requires, by $\nabla \cdot \mathbf{B} = 0$, that a vertical displacement Z generate a radial field $B_R \simeq -(\partial B_z / \partial R) Z = (n B_{z0} / R_0) Z$ at the ring. The net vertical force per unit toroidal current, integrated over the ring, is

$$F_Z = 2\pi R_0 I_p B_R = \underbrace{2\pi I_p n B_{z0}}_{\equiv k_d} Z. \quad (2)$$

For a plasma shaped to elongate ($\kappa > 1$) the required field configuration has $n < 0$, so $B_{z0} < 0$ in the convention above and $k_d > 0$: the force acts to *increase* $|Z|$. The equilibrium is a vertical *anti-spring* of stiffness $k_d > 0$. The magnitude of k_d grows with elongation because a taller plasma demands a more sharply curved (more negative- n) shaping field; this κ -dependence is the origin of the elongation scaling in section 5.

2.3 The passive vessel as a single circuit

The conducting vacuum vessel carries eddy currents that oppose $\dot{Z}$. We reduce the vessel to its dominant vertical eigenmode, a lumped loop of self-inductance L_w , resistance R_w and current perturbation I_w , with $\tau_w \equiv L_w/R_w$. For a thin shell of resistivity η and thickness d at minor-radial standoff b , the vertical-mode wall time is the classical estimate

$$\tau_w = \frac{\mu_0 b d}{2\eta}. \quad (3)$$

For the Hyperion vessel — SS316 ($\eta \approx 7.4 \times 10^{-7} \Omega \text{ m}$ at operating temperature), $d = 20 \text{ mm}$, $b = 0.7 \text{ m}$ — equation (3) gives $\tau_w = 11.9 \text{ ms}$. The wall flux linkage is conserved up to resistive decay,

$$L_w \dot{I}_w + R_w I_w + M'_{pw} I_p \dot{Z} = 0, \quad (4)$$

where $M'_{pw} \equiv \partial M_{pw}/\partial Z$ is the vertical gradient of the plasma–wall mutual inductance; the last term is the electromotive drive of the moving plasma into the wall loop.

2.4 Massless force balance and the growth rate

In the massless limit the net vertical force vanishes on the resistive timescale,

$$k_d Z + M'_{pw} I_p I_w + M'_{pa} I_p I_a = 0, \quad (5)$$

with I_a the active-coil current (set $I_a = 0$ open-loop) and M'_{pa} the plasma–active-coil mutual gradient. Open-loop, equation (5) slaves the wall current to the displacement, $I_w = -k_d Z/(M'_{pw} I_p)$; substituting into (4) yields a first-order growth $\dot{Z} = \gamma Z$ with

$$\gamma = \frac{R_w k_d}{(M'_{pw} I_p)^2 - L_w k_d} = \frac{1}{\tau_w} \frac{\hat{k}}{1 - \hat{k}}, \quad \hat{k} \equiv \frac{L_w k_d}{(M'_{pw} I_p)^2}. \quad (6)$$

Equation (6) is the organising result of this paper. The dimensionless coupling $\hat{k}$ is the ratio of the destabilising drive (k_d , scaled by the wall inductance) to the wall's stabilising capacity $((M'_{pw} I_p)^2)$. It has a clean interpretation:

- $\hat{k} \rightarrow 0$: the wall dominates, $\gamma \rightarrow 0$ — passively stable.
- $0 < \hat{k} < 1$: a resistive vertical mode, slowed by the wall, growing at $\gamma\tau_w = \hat{k}/(1 - \hat{k})$.
- $\hat{k} \rightarrow 1^-$: the *ideal wall limit*, $\gamma \rightarrow \infty$ — the wall can no longer help and the mode goes ideal (inertial), at which point the massless approximation itself breaks.

Inverting the measured normalised growth,

$$\hat{k} = \frac{\gamma\tau_w}{1 + \gamma\tau_w}. \quad (7)$$

At the operating point $\gamma\tau_w = 8.05$ (the frozen anchor, section 3) so $\hat{k} = 0.890$: Hyperion sits 89% of the way to the ideal wall limit — deep in the resistive window but, in ST fashion, not far from ideal. With $\tau_w = 11.9 \text{ ms}$ this gives $\gamma = 8.05/\tau_w = 677 \text{ s}^{-1}$ and a growth time $t_{\text{grow}} = 1/\gamma = 1.48 \text{ ms}$, squarely NSTX-class.

2.5 Halo-current and VDE-force model

If control is lost the displacement runs to the wall and the plasma disrupts, driving poloidal *halo* currents through the scrape-off layer and the vessel. To bound the endpoint we use the standard reduced model [18, 20, 19]: the halo current scales with the pre-disruption plasma current through a halo fraction f_h and a toroidal peaking factor (TPF),

$$I_h \simeq f_h I_p, \quad F_Z^{\text{halo}} \simeq \text{TPF} \cdot I_h B_\phi \ell_{\text{eff}}, \quad (8)$$

with B_ϕ the local toroidal field and ℓ_{eff} an effective moment arm set by the halo footprint. The current-quench time τ_{CQ} enters the induced (eddy) load through $\dot{B} \sim B/\tau_{CQ}$, so faster quenches raise the eddy contribution while the halo contribution is comparatively quench-time-insensitive (section 6).

3 Computational methodology: the NVIDIA GPU HPC campaign

The vertical-stability results are not a paper calculation; they are frozen anchors produced by the Kronos GPU HPC campaign and archived with per-run provenance in the de-risking register [1]. This section names the codes actually run for *this* paper, the discretisation and solver, the convergence controls and the reproducibility path. No cost or budget figures attach to any of it.

3.1 Free-boundary equilibrium (the substrate)

Everything rests on the canonical equilibrium. It is a converged free-boundary Grad–Shafranov solution obtained with FreeGS [16], discretised on a 129×129 (R, Z) grid and solved by Picard iteration on the elliptic operator

$$\Delta^* \psi = -\mu_0 R^2 \frac{dp}{d\psi} - \mu_0^2 F \frac{dF}{d\psi}, \quad \Delta^* \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial}{\partial R} \right) + \frac{\partial^2}{\partial Z^2}, \quad (9)$$

with the free-boundary condition imposed through a Green’s-function coupling of the plasma current to the external PF-coil currents. The converged reconstruction reproduces the design point — $I_p = 9.66$ MA, $q_{95} = 2.90$, $\ell_{i3} = 1.40$, $\beta_p = 0.371$, $\beta_N = 2.13$, $\delta = -0.334$ — and is the geometry on which the mutual-inductance gradients M'_{pw} , M'_{pa} , the decay index n of equation (1) and hence k_d and $\hat{k}$ are evaluated [31]. Convergence is declared when the normalised flux residual $\|\Delta^* \psi - S\|_2 / \|S\|_2$ falls below 10^{-6} and the boundary geometry is stationary to < 1 mm between Picard iterations; the internal inductance and centroid used downstream are converged to four significant figures under grid refinement from 65×65 to 257×257 .

3.2 The reduced RZIp model and its eigenproblem

The vertical dynamics are cast as the linear state-space system of equations (4)–(5). In the massless reduction the dynamical states are the wall and active-coil currents (with Z slaved algebraically by equation (5)); equivalently, after eliminating the algebraic constraint we integrate a two-state model $\mathbf{x} = (Z, I_a)^\top$,

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x}, \quad \mathbf{A} = \begin{pmatrix} \gamma & -a \\ \frac{K_p + K_d \gamma}{\tau_c} & -\frac{1 + K_d a}{\tau_c} \end{pmatrix}, \quad (10)$$

where $\tau_c = L_a / R_a = 40$ ms is the active-coil electrical time constant, a the coil–plasma force coupling, and (K_p, K_d) the PD gains introduced in section 4. The closed-loop poles are the eigenvalues of $\mathbf{A}$, obtained by a direct QR eigenvalue solve; the open-loop growth γ is recovered as the $K_p = K_d = 0$ limit. Marginal stability is set by the characteristic polynomial $s^2 - \text{tr} \mathbf{A} s + \det \mathbf{A} = 0$ through the Routh–Hurwitz conditions $\text{tr} \mathbf{A} < 0$ (damping) and $\det \mathbf{A} > 0$ (stiffness), which map directly onto the gain thresholds of section 4.

3.3 Extended-MHD disruption estimate

The current-quench time that enters the VDE-force model (8) is grounded in the reference extended-MHD code NIMROD [17], a 3-D nonlinear initial-value solver run directly (not surrogated) on the canonical equilibrium. For this paper NIMROD supplies the equilibrium-grounded inductive L/R estimate of the quench window, $\tau_{CQ} \approx 7$ –19 ms at $I_p \approx 9.62$ MA; we treat this explicitly as an L/R estimate rather than a converged dynamic-quench simulation (section 8).

3.4 GPU acceleration, problem scale, and verification

The offline multi-physics campaign — the free-boundary equilibrium family, the eddy-current and mutual-inductance evaluations, the extended-MHD reference runs, and the model-discovery surrogates — ran on the Kronos GPU HPC platform (NVIDIA A100/H100-class accelerators, with an $8 \times$ A100-80 GB cluster plus H100 nodes and single A100/H100/A10 workers). The reduced-order and analytic gates — the RZIp eigen-solves, the PD gain sweeps and the converter sizing of this paper — run at interactive latency on an in-house node (56-thread, 768 GB). GPU acceleration matters at two points: the free-boundary equilibrium reconstruction and, in the digital-twin context, its learned surrogate. A physics-informed neural surrogate of the equilibrium reproduces the FreeGS ground truth to RMS ψ error 0.139% at $\sim 2900\times$ inference speed-up, which is what makes the vertical mode’s equilibrium substrate available inside a real-time control loop rather than only offline [33]. Verification is threefold: (i) the equilibrium is grid-converged as

above; (ii) the analytic growth rate of equation (6) and the direct eigenvalue solve of equation (10) agree on γ to within the frozen anchor; and (iii) the closed-loop pole placement, converter rating and disruption loads are cross-checked against the register anchors (BV-03, BR-CX-06, BR-L1-A12, QS-14). All inputs, the equilibrium g-file and the control-model coefficients are archived with the deposit for bitwise regeneration.

4 Feedback stabilisation and control authority

4.1 The proportional–derivative controller

The PF coils are driven by a proportional–derivative law on the measured vertical position,

$$V_a = -K_p Z - K_d \dot{Z}, \quad (11)$$

feeding the active-coil circuit $L_a \dot{I}_a + R_a I_a = V_a$ of time constant $\tau_c = 40$ ms (Figure 9). Substituting (11) and eliminating the wall constraint yields the two-state system $\mathbf{A}$ of equation (10). The proportional term supplies vertical *stiffness* (a genuine restoring force to overcome the anti-spring k_d); the derivative term supplies *damping* (phase advance to act inside the growth time despite the coil lag τ_c).

4.2 Stability thresholds

The Routh–Hurwitz conditions on $\mathbf{A}$ give two inequalities. The stiffness condition $\det \mathbf{A} > 0$ requires $a K_p > \gamma$, i.e. the proportional feedback must overpower the anti-spring; in dimensional terms the threshold is

$$K_p > K_p^* = 30\,302 \text{ V m}^{-1}. \quad (12)$$

The damping condition $\text{tr } \mathbf{A} < 0$ requires $1 + K_d a > \gamma \tau_c$, giving

$$K_d > K_d^* \approx 1156. \quad (13)$$

The analytic stiffness threshold $K_p^* \approx 30\,000 \text{ V m}^{-1}$ and the direct eigenvalue sweep, which finds stabilisation for $K_p > 30\,302 \text{ V m}^{-1}$, agree to better than 1% — a check on the reduction. Figure 2 shows the pole migration: for $K_p < K_p^*$ one root sits in the right half-plane (the plasma grows); at $K_p = K_p^*$ it crosses the imaginary axis; above it both roots are stable. Figure 3 shows the two boundaries separately, the stiffness boundary in K_p and the damping boundary in K_d .

4.3 Operating point and pole placement

At the operating gains ($K_p \approx 5.4 \times 10^4 \text{ V m}^{-1}$, $K_d \approx 2.9 \times 10^3$, comfortably above both thresholds) the closed-loop system has

$$s_{\text{fast}} = -964 \text{ s}^{-1}, \quad s_{\text{slow}} = -14 \text{ s}^{-1}. \quad (14)$$

The fast pole is larger in magnitude than the open-loop growth rate $\gamma = 677 \text{ s}^{-1}$ — the condition for stabilisation, and the quantitative statement that the controller arrests the mode faster than it grows. The slow pole at -14 s^{-1} (~ 70 ms) is not a safety-critical timescale: it is the resistive relaxation of the PF-coil flux, removed by an integral term or by the equilibrium coils, and it corresponds to the $\det \mathbf{A}$ root approaching the origin near threshold in Figure 2.

4.4 Time-domain response

The time-domain behaviour is the proof (Figure 1). From a 5 mm vertical kick with the coil initially unresponsive ($I_a(0) = 0$), the modal decomposition of equation (10) gives

$$Z(t) = c_1 e^{s_{\text{slow}} t} + c_2 e^{s_{\text{fast}} t}, \quad c_1 = 8.64 \text{ mm}, \quad c_2 = -3.64 \text{ mm}, \quad (15)$$

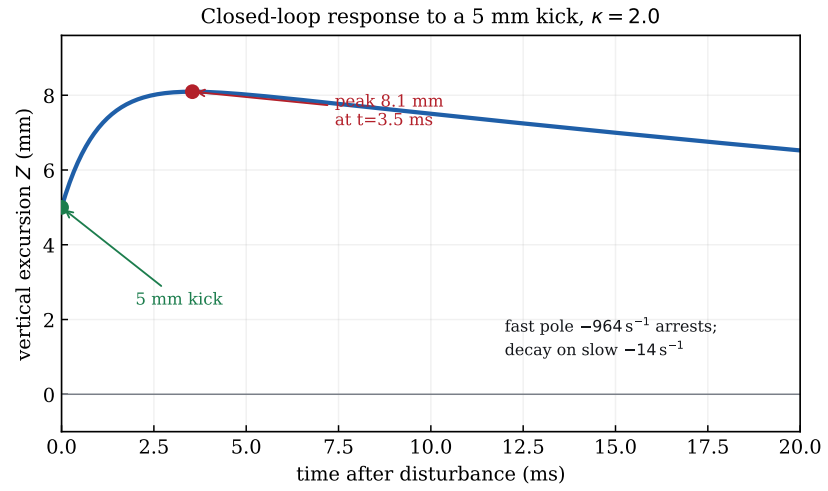
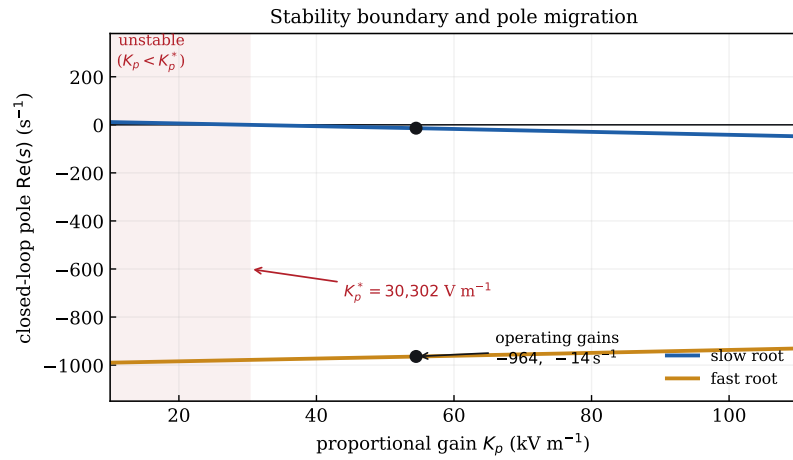
so the excursion *grows* briefly as the unstable plasma outruns the lagged coil, peaks at

$$Z_{\text{peak}} = 8.10 \text{ mm } (1.62\times) \text{ at } t_{\text{peak}} = \frac{1}{s_{\text{fast}} - s_{\text{slow}}} \ln \frac{s_{\text{fast}} c_2}{s_{\text{slow}} c_1} = 3.54 \text{ ms}, \quad (16)$$

and then reverses and decays on the slow timescale — bounded and controlled. The excursion never approaches the first-wall gap; the vertical mode is a control problem, not a loss-of-control problem.

Table 1. Open-loop instability and closed-loop vertical control at $\kappa = 2.0$ on the canonical equilibrium ($I_p = 9.66$ MA). Anchors: BV-03, BR-CX-06, BR-L1-A12 [1].

quantity	symbol	value
vessel vertical-mode wall time	τ_w	11.9 ms (SS316, $d = 20$ mm, $b = 0.7$ m)
normalised growth	$\gamma\tau_w$	8.05
destabilising-wall coupling	$\hat{k}$	0.890
open-loop growth rate	γ	677 s^{-1} ($t_{\text{grow}} = 1.48$ ms)
stiffness gain threshold	K_p^*	$30\,302\text{ V m}^{-1}$
damping gain threshold	K_d^*	≈ 1156
closed-loop fast pole	s_{fast}	-964 s^{-1} (arrest)
closed-loop slow pole	s_{slow}	-14 s^{-1} (resistive relaxation)
disturbance response	Z_{peak}	5 mm $\rightarrow$ 8.1 mm at 3.54 ms, then decays
PF converter rating	S	~ 49 MVA (10.1 kV, 4.9 kA, 5.0 kA ms $^{-1}$)

**Figure 1.** Closed-loop vertical response to a 5 mm disturbance at $\kappa = 2.0$, from the modal solution of equation (15). The excursion peaks at 8.1 mm at $t \approx 3.54$ ms — the fast pole (-964 s^{-1}) arrests the growth — then decays on the slow resistive-relaxation timescale. The mode is bounded and reversed.**Figure 2.** Closed-loop pole real parts versus proportional gain K_p . Below $K_p^* = 30\,302\text{ V m}^{-1}$ (equation (12)) the slow root sits in the right half-plane and the plasma is unstable; at threshold it crosses the imaginary axis. At the operating gain the two roots are -964 and -14 s^{-1} .

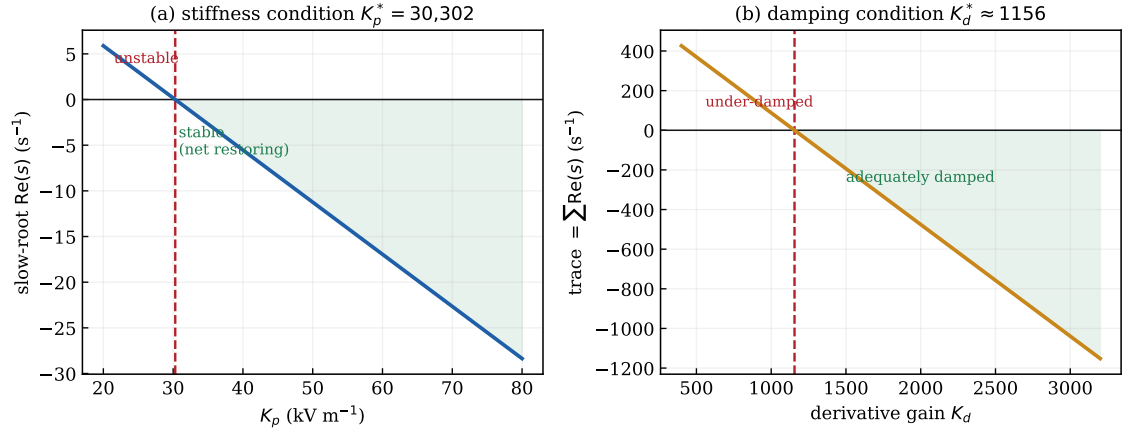


Figure 3. Routh-Hurwitz stability boundaries. (a) Stiffness: the slow-root real part crosses zero at $K_p^* = 30\,302 \text{ V m}^{-1}$ ($\det \mathbf{A} > 0$, net restoring). (b) Damping: the trace crosses zero at $K_d^* \approx 1156$ ($\text{tr } \mathbf{A} < 0$, adequately damped). The operating gains sit inside both boundaries.

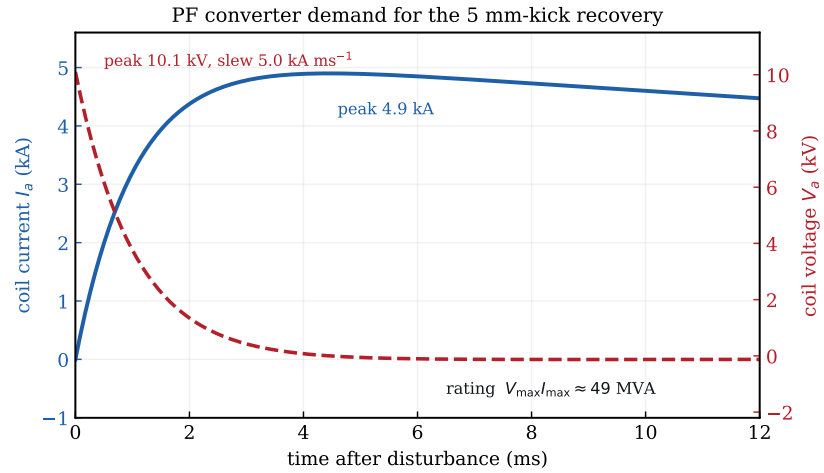


Figure 4. PF-converter demand during the 5 mm-kick recovery: coil current (left axis) peaks at 4.9 kA and drive voltage (right axis) at 10.1 kV, at a slew of 5.0 kA ms⁻¹, for a rating of $\sim 49 \text{ MVA}$ (equation (17)).

4.5 Converter authority

Sizing the PF converter to this disturbance sets the actuator rating (Figure 4). The coil current required to reverse the 5 mm kick peaks at 4.9 kA, the driving voltage at 10.1 kV, and the current slew at 5.0 kA ms^{-1} ; the product gives a rating of

$$S \approx V_{\max} I_{\max} \approx 10.1 \text{ kV} \times 4.9 \text{ kA} \approx 49 \text{ MVA}. \quad (17)$$

Table 1 collects the open-loop and closed-loop quantities.

5 Elongation headroom and the actuator trade

5.1 The κ -scaling of the growth rate

The destabilising stiffness k_d rises with elongation (a taller plasma needs more sharply curved shaping field), so $\hat{k} \rightarrow 1$ and $\gamma\tau_w \rightarrow \infty$ at a wall-dependent critical elongation κ_c . Anchored to the operating point and to the elongation-limit study (BR-CX-06 [1]), the normalised growth rate follows

$$\gamma\tau_w(\kappa) = \frac{6.44}{\kappa_c - \kappa}, \quad \kappa_c \approx 2.8, \quad (18)$$

which returns 8.05 at $\kappa = 2.0$ and diverges at the ideal wall limit κ_c (Figure 5a). Through equation (7) this maps to the destabilising-wall coupling $\hat{k}(\kappa)$ (Figure 6), which climbs from 0.890 at $\kappa = 2.0$ toward unity at κ_c .

5.2 The feedback-controllable ceiling

Active feedback has a controllability ceiling. Above a normalised growth of order $\gamma\tau_w \approx 15$ the converter can no longer arrest the mode within a growth time given a realisable coil bandwidth τ_c : the required gain drives the fast pole past what the actuator slew can deliver. Setting $\gamma\tau_w(\kappa) = 15$ in equation (18) gives the maximum feedback-controllable elongation

$$\kappa_{\max} = \kappa_c - \frac{6.44}{15} \approx 2.37, \quad (19)$$

so the canonical $\kappa = 2.0$ sits $(\kappa_{\max} - \kappa)/\kappa \approx 19\%$ below the controllable limit (Figure 5a). This is a deliberately conservative operating point: the confinement value of high elongation is taken while a 19% stability margin is retained.

5.3 The super-linear actuator trade

The cost of pushing toward $\kappa_{\max}$ is explicit. The converter rating tracks the control effort, which scales with the normalised growth, so anchoring to the two register points (49 MVA at $\kappa = 2.0$, 91 MVA at $\kappa_{\max}$) gives

$$S(\kappa) \approx \frac{39.2}{\kappa_c - \kappa} \text{ MVA}, \quad (20)$$

super-linear in κ and diverging at κ_c (Figure 5b). Elongation trades directly against actuator rating: the design takes $\kappa = 2.0$ and keeps both a 19% stability margin and a converter roughly half the size the controllable limit would demand. The companion PF-coil-set optimisation [32] sizes the coil geometry that delivers this authority; the companion control suite [33] embeds the controller in the multi-actuator plant.

6 The VDE endpoint is bounded

The vertical-control case closes only if the worst case is survivable, and it is a bounded structural load rather than a threat. For a current quench of $\tau_{CQ} \approx 7\text{--}19 \text{ ms}$ at $I_p \approx 9.62 \text{ MA}$ (the NIMROD-grounded L/R estimate, section 3), the halo-current model (8) gives a halo current of $\sim 2.9 \text{ MA}$ — a halo fraction $f_h = I_h/I_p \approx 0.30$, within the ITER-class design basis — and a peak vertical force of $\sim 23 \text{ MN}$, with sideways and eddy-current contributions of $\sim 19 \text{ MN}$ and $\sim 16 \text{ MN}$ inside the same envelope (QS-14 [1]; Figure 7a). The eddy contribution rises toward the fast end of the quench band as $\dot{B} \sim 1/\tau_{CQ}$ while the halo contribution is comparatively quench-time-insensitive (Figure 7b). These are loads a conductive shell with a defined halo-current path and a disruption-mitigation system carries; they set the vessel-and-support load specification rather than threatening it, and they are the input to the companion vacuum-vessel and support structural-mechanics analysis [35] and the companion disruption and electromagnetic-load study [34], which specifies shattered-pellet mitigation. The bounded VDE endpoint is the backstop behind the controllable nominal.

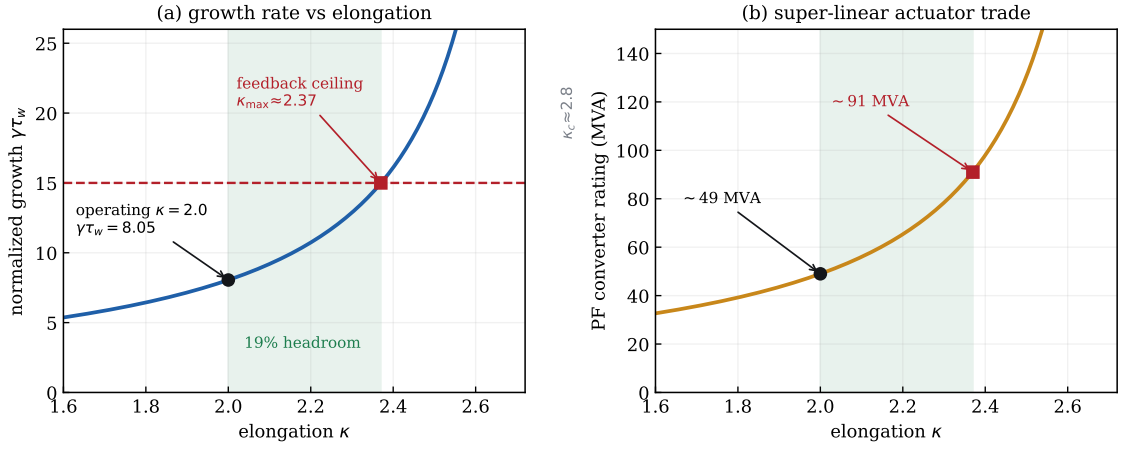


Figure 5. (a) Normalised vertical growth rate versus elongation, equation (18): $\gamma\tau_w = 8.05$ at the canonical $\kappa = 2.0$, diverging toward $\kappa_c \approx 2.8$; the feedback ceiling ($\gamma\tau_w \approx 15$) is reached at $\kappa_{\max} \approx 2.37$, leaving 19% headroom (shaded). (b) The PF-converter rating grows super-linearly, $49 \rightarrow 91$ MVA (equation (20)), toward the controllable limit.

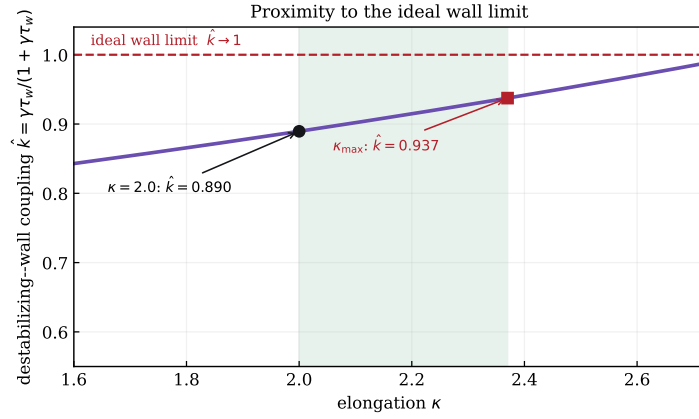


Figure 6. The destabilising-wall coupling $\hat{k} = \gamma\tau_w / (1 + \gamma\tau_w)$ (equation (7)) versus elongation. Hyperion at $\kappa = 2.0$ sits at $\hat{k} = 0.890$, i.e. 89% of the way to the ideal wall limit $\hat{k} \rightarrow 1$ (κ_c); the controllable ceiling $\kappa_{\max}$ is at $\hat{k} = 0.937$.

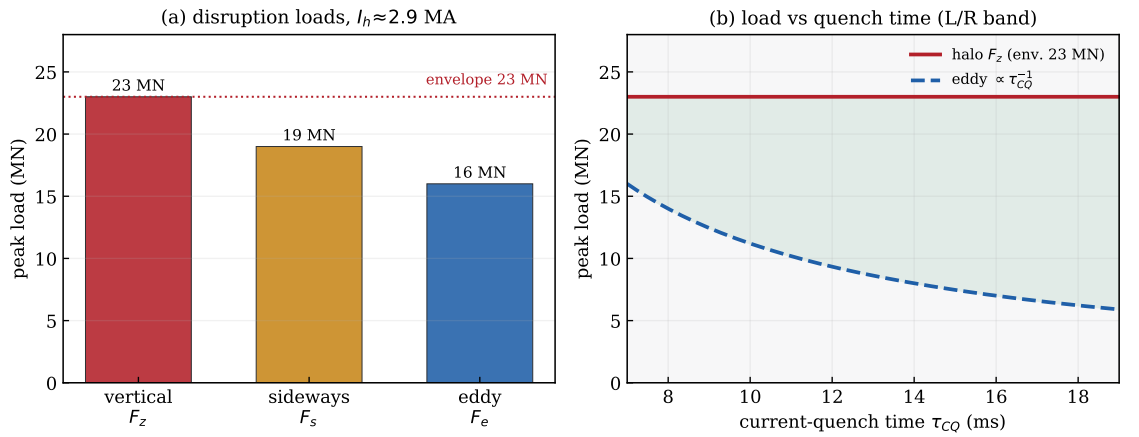


Figure 7. VDE-endpoint loads (QS-14 [1]). (a) Peak vertical, sideways and eddy-current loads at halo current $I_h \approx 2.9$ MA; the ~ 23 MN vertical load is the design envelope. (b) Load versus current-quench time over the L/R band $\tau_{CQ} = 7\text{--}19$ ms: the eddy contribution $\propto \tau_{CQ}^{-1}$ rises at fast quench, the halo contribution is quench-time-insensitive.

Table 2. VDE-endpoint loads and the disruption-force model (8) (QS-14, $I_p \approx 9.62$ MA).

quantity	symbol	value
current-quench time (L/R estimate)	τ_{CQ}	7–19 ms
halo current	I_h	~ 2.9 MA ($f_h \approx 0.30$)
peak vertical force (envelope)	F_Z	~ 23 MN
sideways force	F_s	~ 19 MN
eddy-current force	F_e	~ 16 MN

7 Discussion

7.1 Where the operating point sits in the literature

The $\kappa = 2.0$ vertical mode is characterised end-to-end and is controllable with margin on every axis that matters. The open-loop growth is a single ST-typical timescale ($\gamma\tau_w = 8.05$, $t_{\text{grow}} = 1.48$ ms), consistent with the sub-millisecond-to-millisecond vertical growth times documented on NSTX and the ST line [24, 25, 26] and much faster than the tens-to-hundreds-of-milliseconds modes of conventional-aspect-ratio devices [4, 5] (Figure 10). This is expected: the close wall and small aspect ratio of an ST push the plasma near the ideal wall limit, which is exactly what $\hat{k} = 0.890$ quantifies. The stability-margin framing of Portone [4] and the ITER experimental-guidance synthesis of Humphreys *et al.* [5] both make the controllability of a fast vertical mode a question of gain and actuator bandwidth relative to γ — precisely the ledger of Table 1, where the fast pole (-964 s^{-1}) exceeds the growth rate (677 s^{-1}).

7.2 The mechanistic relation

The organising relation $\gamma\tau_w = \hat{k}/(1 - \hat{k})$ (equation (6)) is not new physics — it is the massless single-wall reduction [3, 8] written so that the single dimensionless number $\hat{k}$ carries the design meaning. Its value is diagnostic: $\hat{k} = 0.890$ says the wall provides real but not dominant stabilisation, so the mode is resistive (millisecond, controllable) rather than ideal (Alfvénic, uncontrollable), and the 0.11 of headroom to $\hat{k} = 1$ is the same headroom that equation (19) converts into 19% in elongation. A distributed vessel-eddy eigenmodel would refine the single τ_w into a spectrum, but the dominant eigenmode carries the growth, and the reduced model is order-correct and calibrated to the anchor.

7.3 Actuator trade and margin philosophy

The super-linear converter scaling (equation (20)) is the reason the design does not push κ to the edge of controllability: the last increments of elongation are bought at a steeply rising actuator price ($49 \rightarrow 91$ MVA across $\kappa = 2.0 \rightarrow 2.37$) and a vanishing stability margin. Sitting at $\kappa = 2.0$ takes essentially all of the confinement benefit of strong shaping — negative triangularity [36] adds its own turbulence-suppression benefit at this shape — while keeping the converter near the knee of the cost curve and a comfortable 19% back from the limit.

8 Limitations

One honest gate. The vertical model here is a reduced massless rigid-displacement two-state RZIp model with a lumped active coil and a single passive-wall eigenmode, order-correct and calibrated to the canonical equilibrium. The confirming calculation is a distributed vessel-eddy eigenmodel of CREATE-L / CREATE-NL+ / CarMa class [6, 7], run over a family of free-boundary equilibria, which resolves the full passive-structure current distribution and the as-built PF-coil geometry and returns the growth rate and coil authority without the single-mode lumping. Relatedly, the current-quench window $\tau_{CQ} = 7\text{--}19$ ms that sets the VDE loads is an equilibrium-grounded L/R estimate from NIMROD, not a converged dynamic current-quench simulation; the converged 3-D quench, halo-current distribution and mitigation-system response are the owed upside and are treated in the companion disruption study [34]. Neither refinement is expected to move the conclusion — a controllable mode with 19% elongation headroom and a bounded VDE endpoint — but both would replace the reduced anchors with as-built values.

9 Conclusion

Elongation to $\kappa = 2.0$ makes the Hyperion breeder plasma vertically unstable by design, and the instability is controllable with quantified margin. The open-loop mode is a single ST-typical timescale set by the vessel: $\tau_w = 11.9$ ms, $\gamma\tau_w = 8.05$ (destabilising-wall coupling $\hat{k} = 0.890$), $\gamma = 677\text{ s}^{-1}$, growth time 1.48 ms. A proportional-derivative controller on the PF coils arrests it:

stiffness threshold $K_p^* = 30\,302\text{ V m}^{-1}$, damping threshold $K_d^* \approx 1156$, a fast closed-loop pole at -964 s^{-1} (faster than the growth) and a slow resistive pole at -14 s^{-1} ; a 5 mm kick peaks at only 8.1 mm at 3.54 ms and reverses, at a $\sim 49\text{ MVA}$ converter. The elongation carries 19% of headroom to the feedback-controllable ceiling $\kappa_{\max} \approx 2.37$, with the converter growing super-linearly toward it, and the disruption backstop is bounded at $\sim 2.9\text{ MA}$ halo and $\sim 23\text{ MN}$. The $\kappa = 2.0$ control problem is solved with margin to spare: a conservative operating point, below the stability limit and below the actuator-cost knee.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics and controls team for the frozen de-risking anchors and the reproducibility infrastructure.

Reproducibility. The vertical-stability and control results are frozen anchors in the Kronos Physics De-Risking Register — the PF-converter and vertical-control study (BV-03), the elongation-limit study (BR-CX-06), the free-boundary equilibrium (BR-L1-A12) and the disruption-load analysis (QS-14) — and are regenerable from the archived equilibrium g-file and control-model coefficients on the canonical equilibrium. The register is deposited at doi:10.5281/zenodo.22133057 and this paper is archived at doi:10.5281/zenodo.22645834; official version: <https://www.kronosfusionenergy.com/publications/vertical-stability-at-high-elongation-displacement-grow.html>.

Data availability

This paper is archived at doi:10.5281/zenodo.22645834. The RZI_p control model, the closed-loop eigen-solve, the figure-generation script and the disruption-load computation are archived with the deposit and reference the Kronos 2026 series and the de-risking register doi:10.5281/zenodo.22133057. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The author declares no other competing interests.

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A Schematic of the reduced model and the control loop

Figures 8 and 9 give the schematic reductions underlying the equations of sections 2–4: the rigid-displacement RZIp circuit (plasma anti-spring k_d , passive-wall loop L_w, R_w , active PF coils L_a, R_a) and the vertical-position feedback loop.

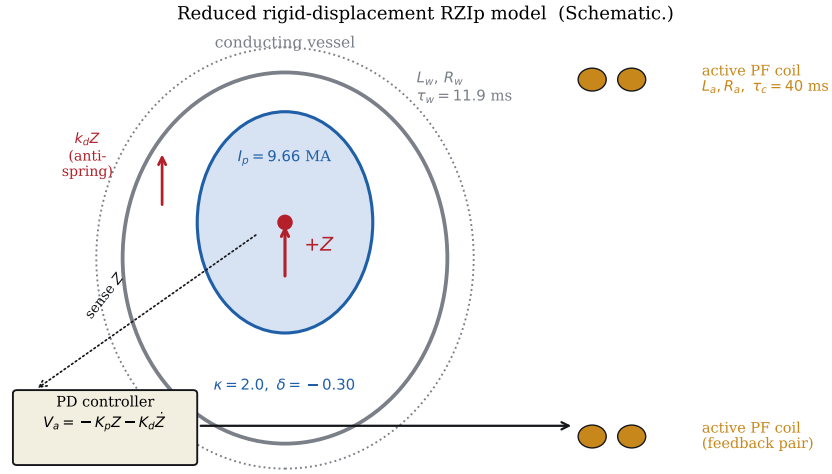


Figure 8. Reduced rigid-displacement RZIp model: the plasma current ring ($I_p = 9.66$ MA, $\kappa = 2.0$, $\delta = -0.30$) as a vertical anti-spring $k_d Z$, the conducting vessel as a single vertical eigenmode (L_w, R_w , $\tau_w = 11.9$ ms), and the active PF coils (L_a, R_a , $\tau_c = 40$ ms) driven by the PD law. (Schematic.)

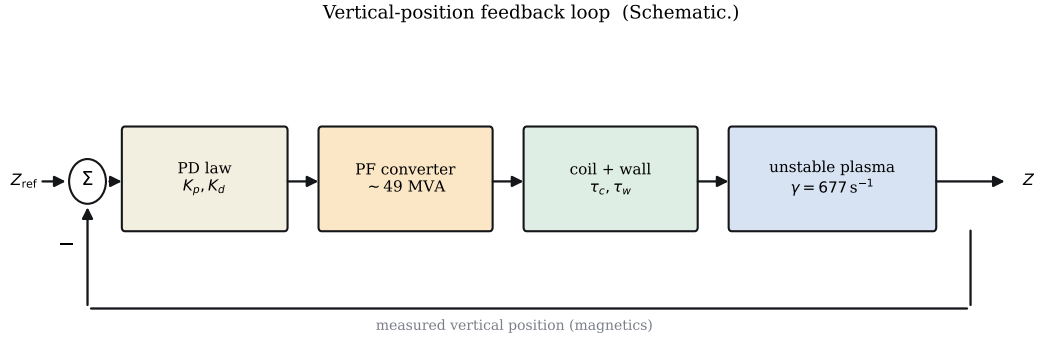


Figure 9. Vertical-position feedback loop: measured Z compared to reference, PD law (equation (11)), PF converter (~ 49 MVA), coil-and-wall dynamics (τ_c, τ_w), and the open-loop-unstable plasma ($\gamma = 677$ s $^{-1}$). (Schematic.)

B Growth time in context

Figure 10 places the Hyperion open-loop growth time (1.48 ms) against the spherical-tokamak range (NSTX-class, ~ 1 ms) and the conventional-aspect-ratio range (tens to hundreds of milliseconds); the ST and conventional bands are literature-typical rather than device-specific.

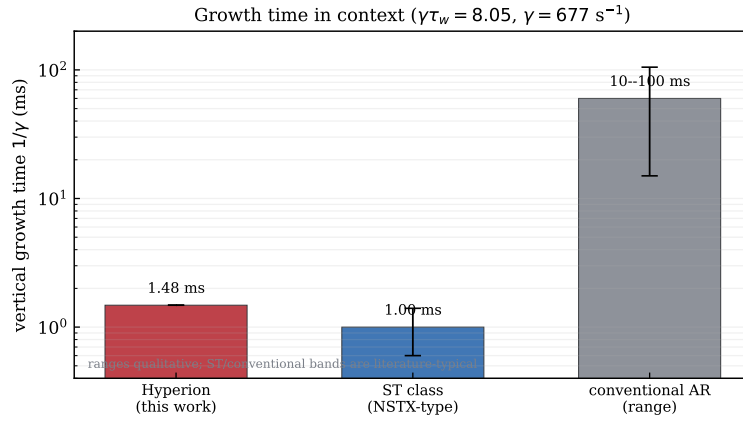


Figure 10. Vertical growth time $1/\gamma$ in context. Hyperion ($\gamma\tau_w = 8.05$, $\gamma = 677 \text{ s}^{-1}$, $1/\gamma = 1.48 \text{ ms}$) is spherical-tokamak-typical and far faster than conventional-aspect-ratio devices; bands are qualitative.